

Evaluation of load-carrying capacity of horizontal joints in concrete wind turbine towers

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ABSTRACT

Concrete wind turbine towers are typically subjected to complex combined loading, including compression, bending, shear, and torsion, due to wind actions and prestressing forces. However, current design methodologies are inadequate for accurately assessing the load-carrying capacity of horizontal joints under such multi-axial stress conditions. To accurately evaluate the load-carrying capacity of horizontal joints in concrete wind turbine towers subjected to combined compression, bending, shear, and torsion, three scaled-down concrete tower specimens were tested under representative multi-axial loading conditions. The failure modes, load-displacement responses, and other key parameters were measured and analysed. The results demonstrated that the torsional and shear resistance of the horizontal joint regions was primarily governed by the frictional force developed at the joint interface as a result of prestressing. While higher prestress level enhanced the ultimate load-carrying capacity, they also led to a reduction in the ductility of the specimens. A refined finite element (FE) model was subsequently developed in ABAQUS and calibrated against the experimental results. Building upon established calculation methods for evaluating the torsional and flexural capacities of horizontal joints in concrete towers, and based on the fundamental form of the interaction equation for structural load-carrying capacity under combined loads, a new analytical framework was established to evaluate the load-carrying capacity of horizontal joints under combined loading conditions. Based on the refined FE model, a series of parametric numerical analyses were performed. Subsequently, the coefficients in the proposed evaluation equations were calibrated using the numerical simulation results, leading to the development of a comprehensive method for evaluating the load-carrying capacity of horizontal joints.

1. Introduction

The rapid pace of global economic development has led to a continuous increase in energy demand, giving rise to critical challenges such as climate change and energy shortage. As a renewable resource with substantial potential, wind energy has emerged as a key solution for mitigating these global issues [1,2]. To reduce the levelized cost of electricity and enhance the energy output of wind turbines, the rotor diameter and hub height of onshore wind turbines have significantly increased in recent years. As a result, steel-concrete hybrid towers, which offer higher load-carrying capacity, improved lateral stiffness, and lower construction and maintenance costs compared to conventional steel towers, have become the preferred structural system for onshore wind turbines with hub heights exceeding 140 m [3–6].

To address the limitations of traditional cast-in-place construction methods - such as high construction complexity, extended construction durations and difficulties in quality control, prefabricated pipe segment structures seem to be the primary form of concrete wind turbine towers. These prefabricated segments are manufactured in controlled environments and transported to the site for assembly, which significantly enhances both construction efficiency and overall structural capacity. Specifically, the vertical joints - the connections between adjacent pipe segments - are assembled using arc-shaped bolts and epoxy resin to form complete concrete tower segments. The horizontal joints, which connect these concrete tower segments, are bonded with epoxy resin [7,8]. In addition, the concrete tower is anchored to the foundation and transition section through the installation of external prestressed tendons. A schematic illustration of a typical concrete wind turbine tower is shown

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in Fig. 1.

Under the combined effects of wind loads, prestressing forces, and the self-weight of the superstructure, the towers are subjected to a combination of compression, bending, shear, and torsion. It is particularly noteworthy that, under ultimate load conditions, weak connection regions - such as horizontal joints - are prone to cracking. This cracking alters the flexural and torsional section moduli of the joint cross-section, rendering it inconsistent with traditional calculation models and evaluation methods based on the thin-walled member theory. In hybrid wind turbine towers, the concrete tower serves as the sole vertical bearing component. The accuracy of evaluating the load-carrying capacity of weak connection regions is critical to ensuring the structural safety and reliable operation of wind turbines. Therefore, it is essential to investigate the load-carrying capacity of horizontal joints in concrete tower structures. [9–11].

In recent years, the load-carrying capacity of horizontal joints in concrete wind turbine towers has been systematically investigated through theoretical analysis, numerical simulation, and experimental verification. In terms of theoretical modelling, Hartwig and Marx [12] pointed out that the classical Saint-Venant torsion theory fails to accurately describe the torsional behaviour of modern conical towers, and therefore proposed a modified model that accounts for the influence of bending moments. Subsequently, Hartwig and Middendorf [13] developed a torsional truss model capable of capturing the interaction among bending, shear, and torsion. Following this research direction, Chen et al. [14] assumed that failure initiates at horizontal joints and derived an integral expression for the ultimate torsional capacity under combined loading, validating the model through Python-based numerical implementation and small-scale experimental tests. At the design and code level, *T/CEC 5008* [15] introduced a torsional capacity formulation for tower cross-sections, derived from classical mechanics principles and considering the weakening effect of shear force on torsional resistance; Kang et al. [16] derived and compared the torsional design provisions in *DIN EN 1992-1-1* [17], and pointed out that the St. Venant's approach significantly underestimate the actual capacity once joint opening occurs. Paredes and Barbat [18] showed through numerical analyses that although local damage models have limited effects on global stiffness, they considerably alter the damage distribution. Moreover, the experimental and parametric finite element studies by Ren et al. [19], and

Fürll et al. [20] further revealed that existing calculation methods exhibit large uncertainties under combined compression-bending-torsion conditions, and that increasing bending moments continuously weaken the torsional capacity. On this basis, Tan et al. [21] proposed a new compression-bending capacity evaluation approach, whose reliability was confirmed through both experiments and numerical simulations.

However, based on reviewing the existing methods for assessing the load-carrying capacity of horizontal joints in concrete wind turbine towers, several limitations are found: (a) limited research on the load-carrying capacity of towers with cracks on horizontal joints due to bending at ultimate, where the flexural and torsional section properties and the stress distribution along the cross-section of the joint would change significantly; (b) Although some scholars have studied the bearing capacity of horizontal joints under combined loads, the indirect interaction between various loads is typically analysed through physical mechanisms, and the proposed methods often rely on multi-step iterative calculations. Therefore, it is essential to develop an engineering algorithm for evaluating the bearing capacity of horizontal joints in wind turbine towers under combined loads, which should account for cracking at the joints and reflect the direct interaction effects between different loads.

This study includes experimental programme, numerical simulation and design proposal. In the experimental programme, three concrete tower specimens subjected to such combined loads were tested. Load-displacement responses, failure modes, bearing capacities, strain measurements during loading, and variations in prestress were recorded for each specimen. Subsequently, FE model validation was performed using ABAQUS. Building on the validated modelling conditions, an extensive numerical simulation was conducted. Based on the parametric study, a new evaluation approach by integrating the torsional resistance formulation for horizontal joints in concrete towers specified in *DIN EN 1992-1-1:2011-01* [17] and the flexural capacity method proposed by Tan et al. [21]. The newly proposed design method was finally validated by a large number of numerical simulation results.

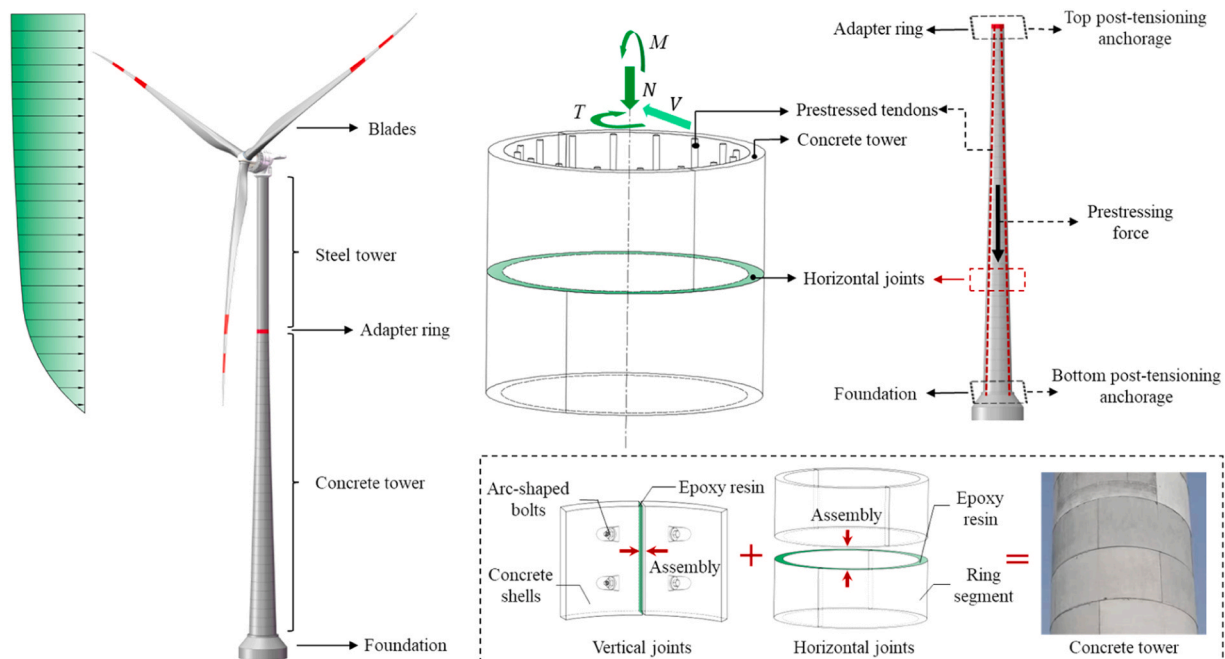


Fig. 1. Schematic drawing of the steel-concrete hybrid tower.

2. Experimental programme

2.1. Specimens

A total of three reinforced concrete tower specimens were designed for this study, designated CT-1 to CT-3. Each specimen was composed of a lower tower segment, an upper tower segment, and a transition segment, as shown in Fig. 2. Compared to full-scale concrete wind turbine towers, specimens CT-1 and CT-2 represent approximately a 1:10 scale, while specimen CT-3 corresponds to a 1:4 scale. The wall thickness was uniformly set at 75 mm to ensure the reinforcement detailing closely matched that of the actual concrete wind turbine towers. The horizontal joint was located at the interface between the upper and lower tower segments. Given that epoxy resin primarily serves as a levelling agent, it was not used in horizontal joints of this study as the levelling has been precisely controlled in the lab. To ensure that the prestress level in the specimen matched that of the actual tower structure, the applied prestress load was controlled through a precompression ratio, which comprehensively accounted for the geometric configuration and material properties of the tower section. The precompression ratio was determined according to Eq. (1). Detailed design parameters for each specimen are listed in Table 1. However, given the limitations of the test site and loading devices as well as the safety of the personnel, the precompression ratio of the large-scale specimen CT-3 was set to 0.2.

$$\alpha_{py} = \frac{F_{py}}{f_{ck} \bullet A} \quad (1)$$

where F_{py} represents the resultant prestressing force applied to the specimen; f_{ck} denotes the characteristic compressive strength of the concrete; A indicates the total cross-sectional area at the horizontal joint.

Connecting discs were installed at the top of the upper tower section, the bottom of the lower tower section, and both ends of the transition section. Axial compression tests on standard concrete cube specimens yielded an average compressive strength of 82.3 MPa. Tensile tests on the reinforcing steel bars indicated an average yield strength of 447 MPa and an ultimate tensile strength of 560 MPa. The reinforcement layouts of specimens are shown in Fig. 3. The reinforcement of the specimens was arranged in accordance with the reinforcement layout and reinforcement ratio of the actual concrete wind turbine towers. Given that the bottom cantilevered section was intended to be connected to the support, the reinforcement in this region was densified. The primary dimensions of specimen CT-3 are indicated in parentheses within the figure. The fabrication process of the specimens primarily involved reinforcement cage assembly, formwork installation, and concrete casting and curing [22], as illustrated in Fig. 4. The levelness and flatness of the baseplate were controlled with the aid of a spirit level and a laser level. Furthermore, sufficient vibration was applied during casting to enhance specimen quality and the smoothness of the joint interface, thereby ensuring effective and smooth load transfer during the subsequent loading phase.

2.2. Test setup

The test was performed using a 100-ton hydraulic servo actuator, and the test setup is shown in Fig. 5. The top and bottom ends of the specimen were connected to the loading beam and the foundation, respectively, using high-strength bolts, ensuring a reliable connection between the specimen and the ground, as well as proper attachment to the actuator for load application. Additionally, the transition section was bolted to the upper tower segment using high-strength bolts (as shown in Fig. 5(a)). Prior to specimen assembly, minor concrete flash and debris around the horizontal joint section were removed by grinding treatment, and the alignment accuracy of the upper and lower segments was controlled within 2 mm in this study. The upper and lower segments were post-tensioned using eight external prestressed tendons anchored at the top of the loading beam and the base of the foundation, respectively. Load cells were installed at the foundation anchorage to monitor the prestressing force in real time during both the tensioning and subsequent loading stages. A single-end tensioning scheme was adopted, with the active anchorage located at the top of the loading beam. (as shown in Fig. 5(b)). After the prestress was applied, the horizontal joints were measured using a vernier calliper to ensure that the local gap width at the joint edges did not exceed 0.5 mm, thereby ensuring effective and smooth load transfer. The foundation was anchored to the ground using anchor bolts. Both ends of the hydraulic actuator were connected to the loading beam and the reaction frame via high-strength bolts, while the reaction frame itself was secured to the reaction wall using additional anchor bolts. The schematic diagrams in Fig. 5 illustrate the loading setup for small-scale specimens. For specimen CT-3, the actuator was positioned on the opposite side of the specimen due to space constraints in the laboratory, and the number of external prestressed tendons was increased to 16; all other configurations remained unchanged.

2.3. Measurement scheme

During the test, primary measurements were conducted on the concrete strain and displacement at the horizontal joint sections, as shown in Fig. 6. Eight strain gauges were installed around the horizontal joint to monitor concrete strains. Additionally, one displacement transducer was placed in each of the four cardinal directions (east, south, west, and north) to record vertical displacements at the joint interface. Additionally, two displacement transducers (labelled DS-5 and DS-6) were installed at the elevation of the loading beam to measure the actuator displacement and the lateral displacement at the specimen's top, respectively.

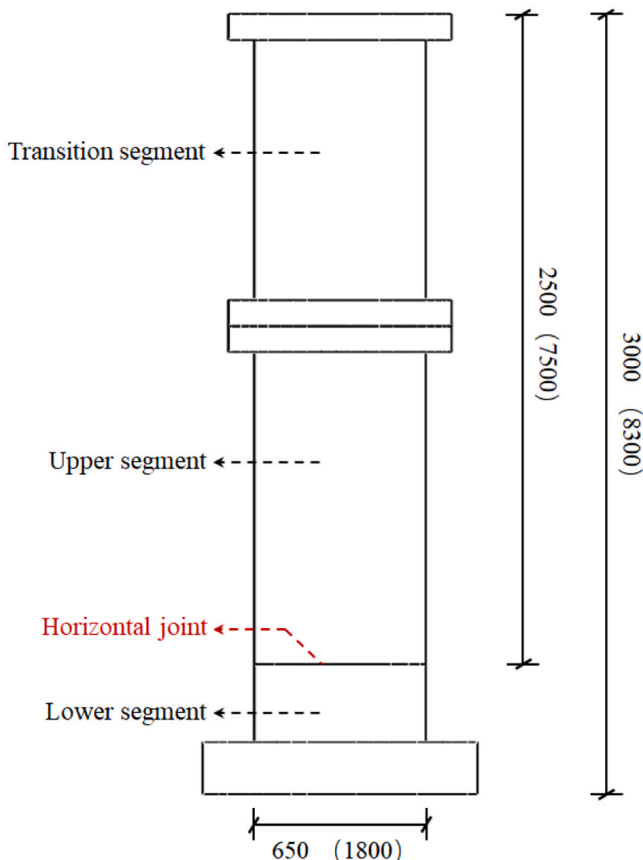
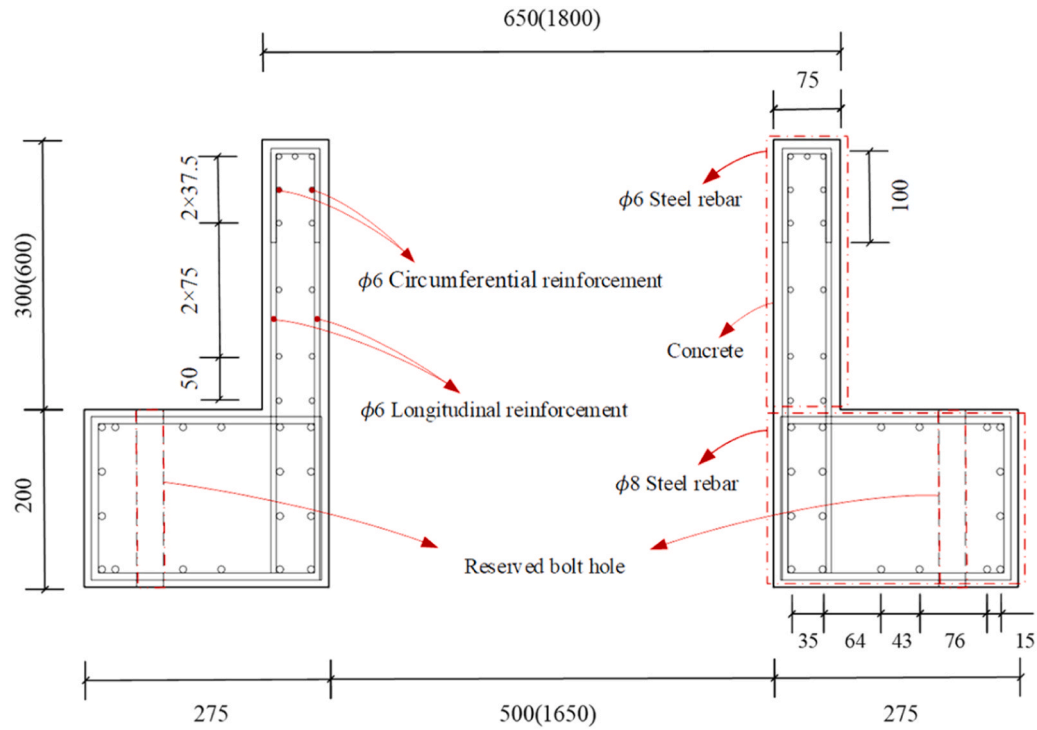


Fig. 2. Schematic of specimens (Unit: mm).

Table 1

Test parameters of specimens.

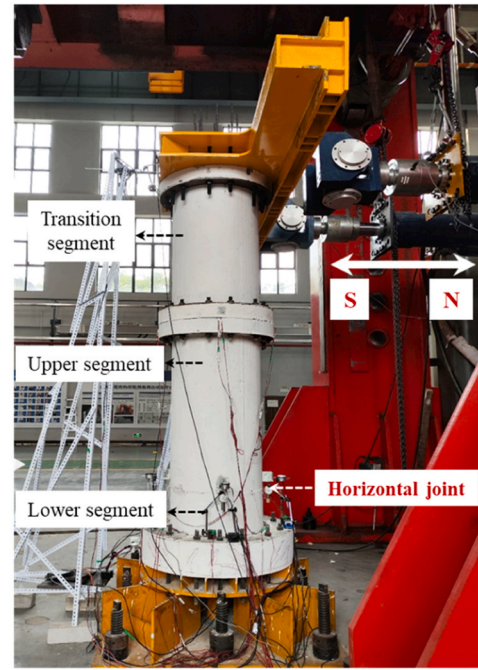
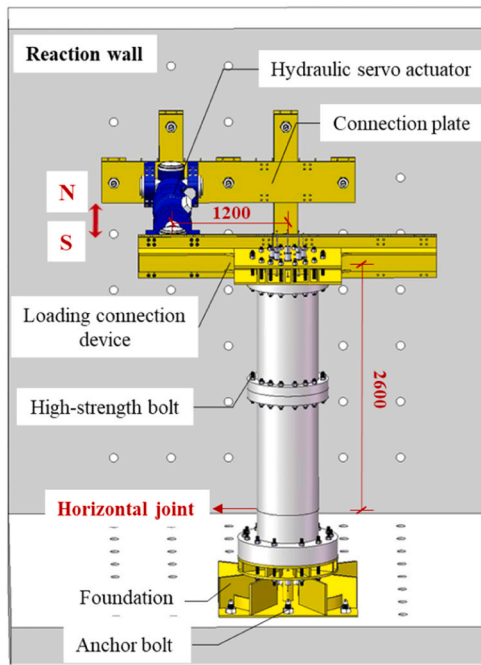
Specimen number	Concrete grade	reinforcement grade	Total height (mm)	Outer diameter (mm)	Wall thickness (mm)	Precompression ratio
CT-1	C60	HRB400	3000	650	75	0.3
CT-2	C60	HRB400	3000	650	75	0.15
CT-3	C60	HRB400	8300	1800	75	0.2

**Fig. 3.** Reinforcement configurations of specimens (Unit: mm).**a. Rebar building****b. Formwork
installation****c. Concrete casting****d. Curing****Fig. 4.** Fabrication process of the specimens.

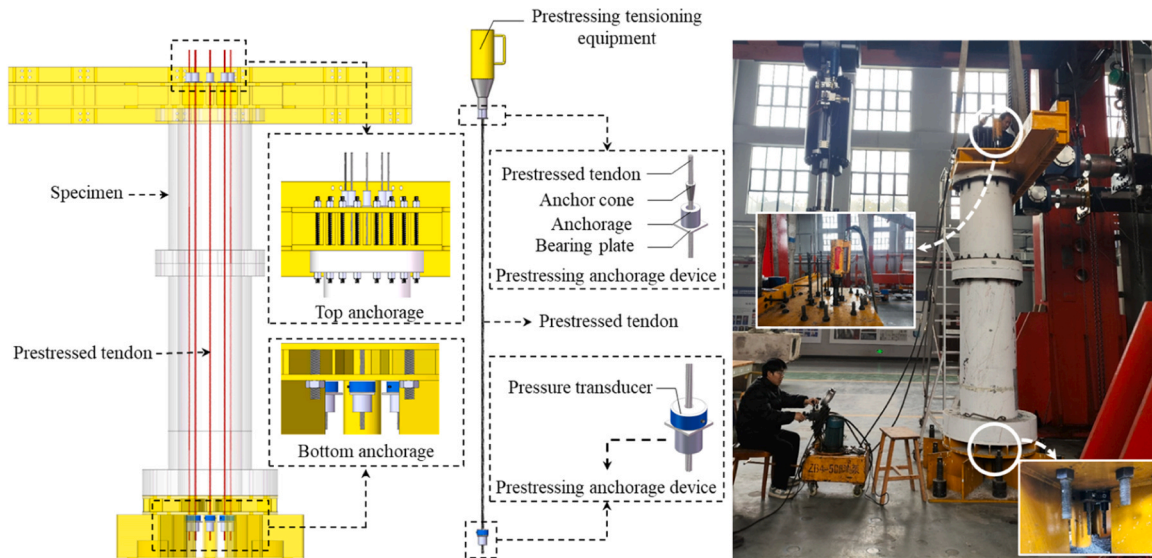
2.4. Loading procedure

To simulate the actual stress state of concrete wind turbine towers, horizontal lateral loads were applied to the specimens using a hydraulic servo actuator under displacement-controlled loading, so as to achieve the effect of simultaneously applying bending moment and torque. For

the small-scale specimens, each loading step adopted a displacement increment of $\Delta y = 2$ mm, with a constant loading rate of 0.2 mm/s. Each loading cycle involved increasing the displacement to the target value and unloading to zero. During the loading process, the failure condition of the specimen and measurements from all displacement transducers, strain gauges, and anchor load cells were recorded at each loading step.



(a) Overall diagram (Unit: mm)



(b) Prestressed tendons loading diagram

Fig. 5. Test setup.

Testing was terminated when the specimen's load-carrying capacity dropped to 85% or less of its peak load. The loading procedure for the large-scale specimen was similar to that of the small-scale specimens, where the displacement increment and loading rate were modified to 5 mm and 0.5 mm/s, respectively. The loading procedure is shown in Fig. 7.

3. Test result analysis

3.1. Phenomena and failure modes

Specimen CT-1 exhibited a combined torsion-bending failure mode, as shown in Fig. 8. On the compression side (north side), numerous

torsional cracks developed at an angle of approximately 75° to the horizontal, while on the tension side (south side), relatively scattered cracks appeared at an angle of approximately 45° to the horizontal. Following the formation of extensive torsional cracking, the surface concrete around the horizontal joint in the compression zone was crushed and subsequently spalled. In contrast, Specimens CT-2 and CT-3 displayed simpler failure modes, both dominated by compression failure, as shown in Figs. 9, 10. In Specimen CT-2, large-scale concrete crushing occurred around the horizontal joint within the compression zone, accompanied by only a few torsional cracks in the upper tower segment. Specimen CT-3 exhibited a smaller but continuous crushing region near the horizontal joint, extending to nearly half the circumference and covering almost the entire compression zone. Several

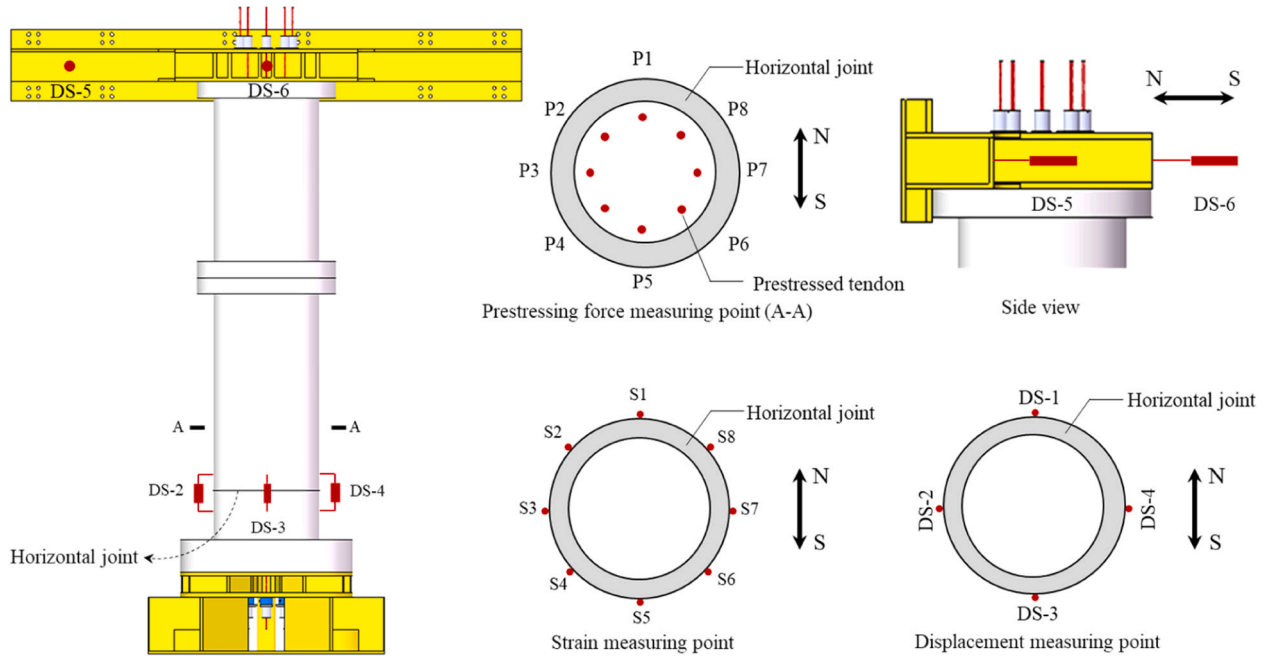


Fig. 6. Measurement scheme.

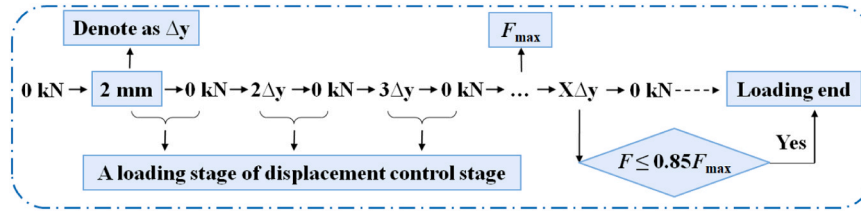


Fig. 7. Loading procedure.

torsional cracks were observed; however, most were confined to the vicinity of the horizontal joint and did not propagate significantly.

Since only prestressed control specimens were designed, only a preliminary discussion on the effect of prestressing is presented. Specimen CT-1 has a relatively high prestress level (precompression ratio of 0.3). According to Coulomb's law, the shear strength of the horizontal joint is correspondingly high. During the intermediate and later loading stages, the shear stress induced by torsion becomes increasingly significant. The combined action of this shear stress and the normal stress caused by bending leads to the formation of torsional cracks in the tower. Meanwhile, as the bending moment arm is larger than the torsional moment arm, the final failure mode is still governed by local crushing of concrete in the compression zone, and the angle of torsional cracks in the tower generally exceeds 45° . In contrast, specimen CT-2 has a lower prestress level (precompression ratio of 0.15), and the torsion-induced shear stress is relatively small. Consequently, torsional slip of the upper tower segment becomes more pronounced during loading. The principal tensile stress resulting from the combined action of shear and normal stresses is lower than the tensile strength of concrete; therefore, almost no torsional inclined cracks develop in the tower. The failure characteristics of specimen CT-3 indicate that concrete crushing predominantly occurs in the compression zone of the horizontal joint near the loading point. It should be noted that other factors, such as joint non-uniformity and concrete inhomogeneity, may cause non-uniform stress distribution within the cross-section and the presence of weak regions in the specimens, potentially leading to minor variations in failure locations and degrees. However, strict control was maintained during the early stages of specimen preparation and during

the installation of the test setup. Therefore, the effects of specimen defects or material inhomogeneity are not further considered in the analysis of the test results.

3.2. Load-displacement curves

The horizontal load-displacement ($F-\Delta$) hysteresis curves of the specimens are shown in Fig. 11. The load corresponds to the data monitored by the actuator's internal transducer, while the displacement refers to the data measured by displacement transducer DS-5. Specimens CT-1 and CT-3 exhibited a significant drop in load-carrying capacity following the peak load, indicative of brittle failure. In contrast, specimen CT-2 showed a clear yielding phase, and the degradation of its load-carrying capacity after reaching the peak was relatively gradual, reflecting a ductile failure behaviour. Additionally, the ultimate load-carrying capacity of specimen CT-1 is 70.9% higher than that of specimen CT-2. This can be attributed to the higher prestress level, which increases friction at the horizontal joint interface, thereby enhancing the specimen's torsional and shear resistance. At the same time, the bending moment required to initiate cracking at the horizontal joint is elevated. These combined effects contribute to a notable improvement in the specimen's ultimate load-carrying capacity. At lower prestress level, the bending moment required to initiate separation at the horizontal joint is reduced, resulting in diminished overall integrity and lower frictional resistance along the joint interface. Nevertheless, the decreased prestress enhances the deformation capacity of the specimen, leading to significantly greater ductility in specimen CT-2 compared to specimen CT-1.

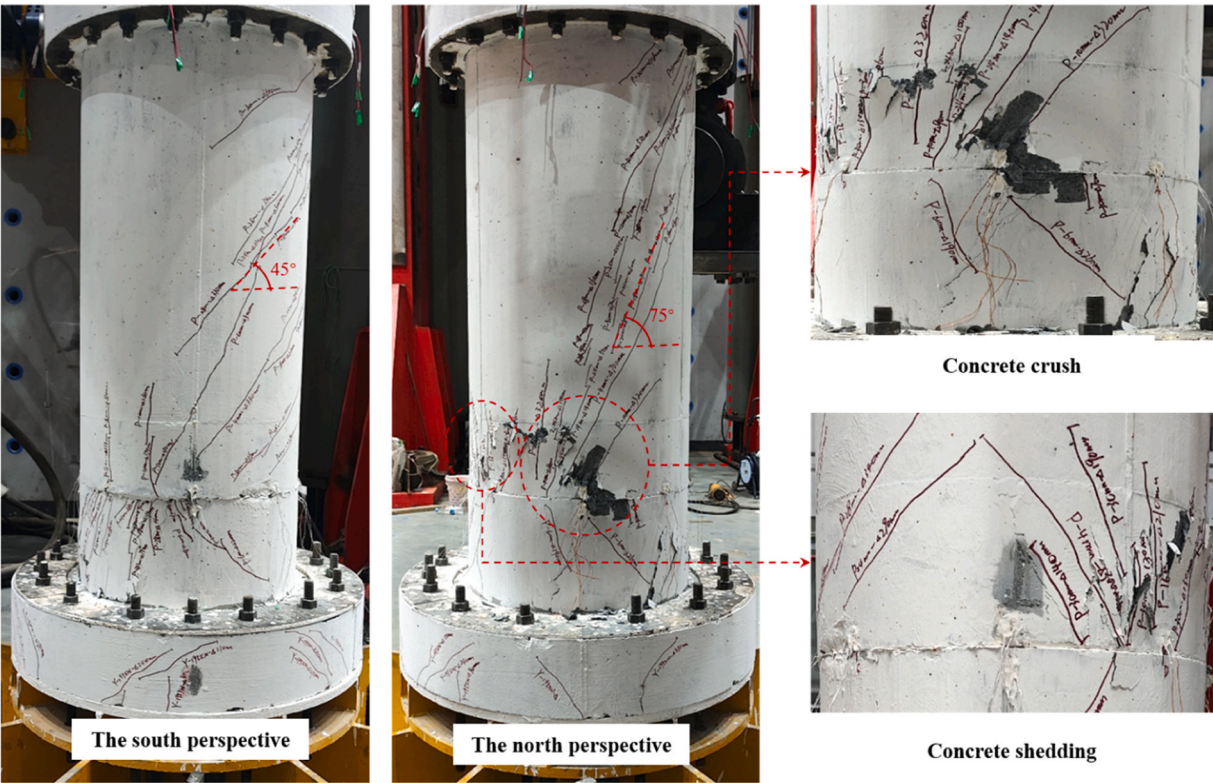


Fig. 8. failure mode of specimen CT-1.

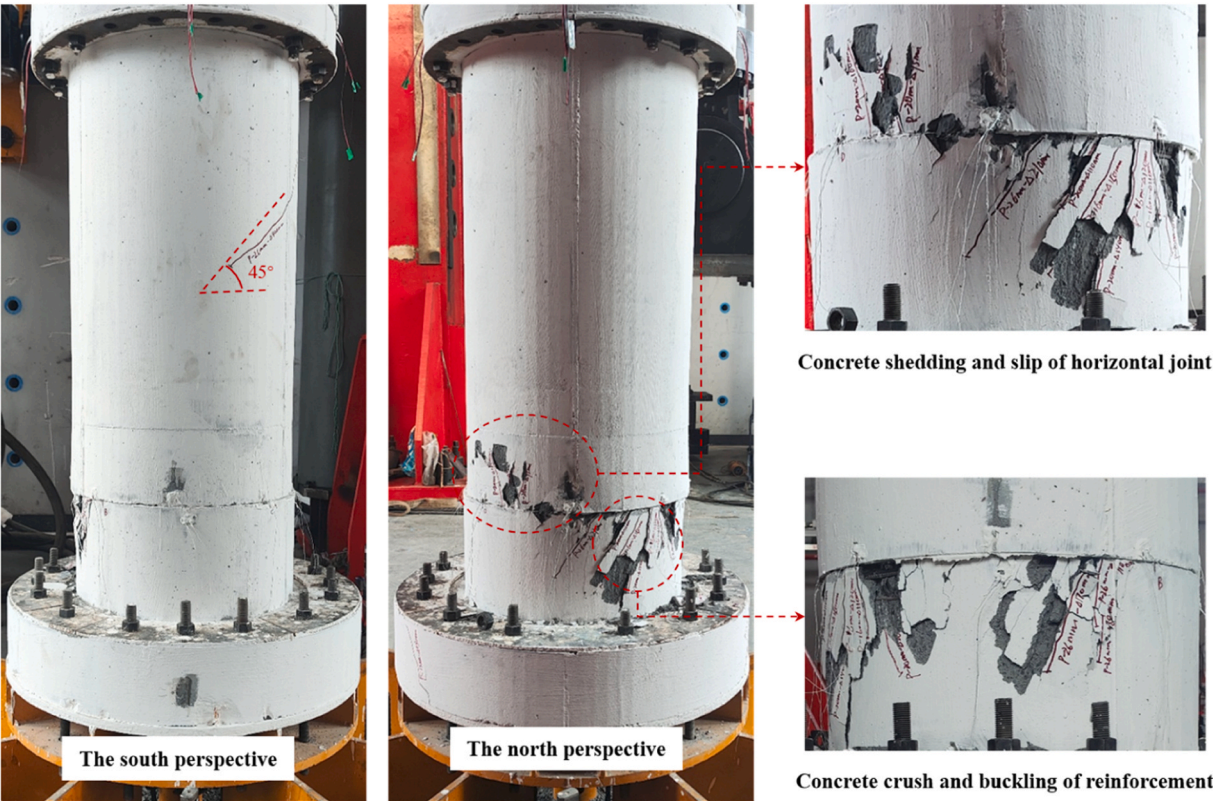


Fig. 9. Failure mode of specimen CT-2.

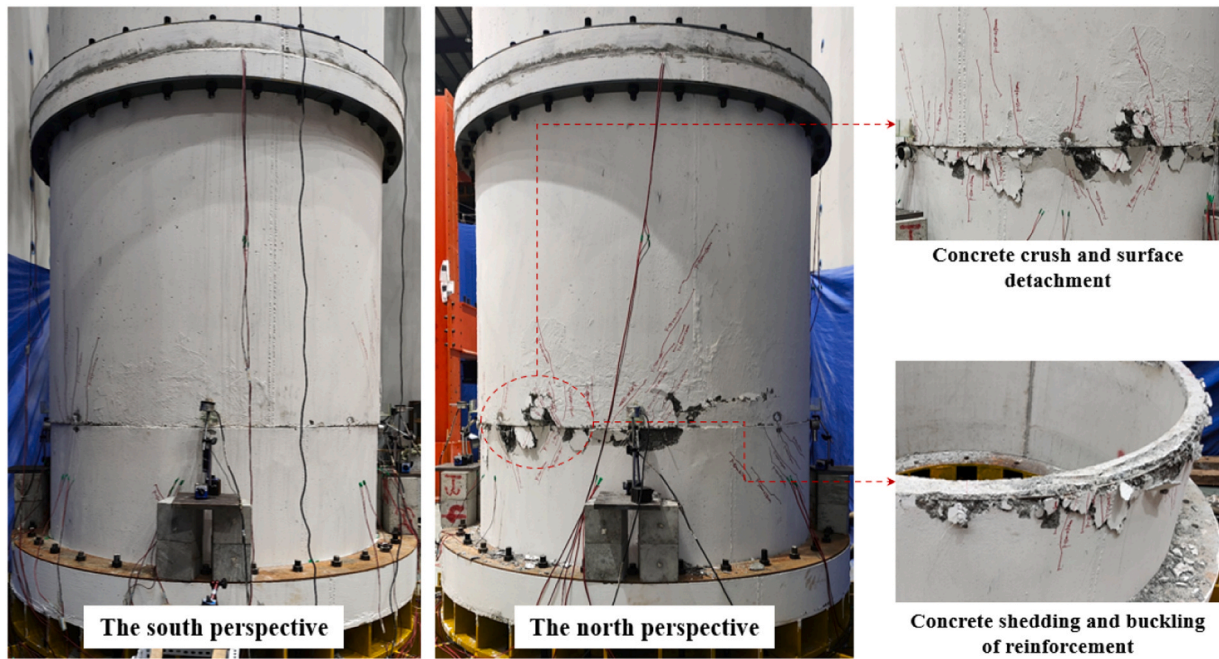
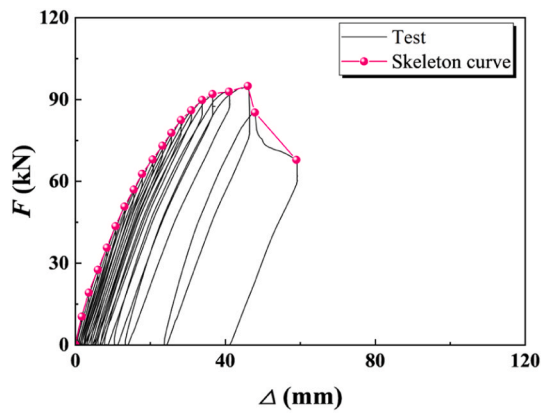
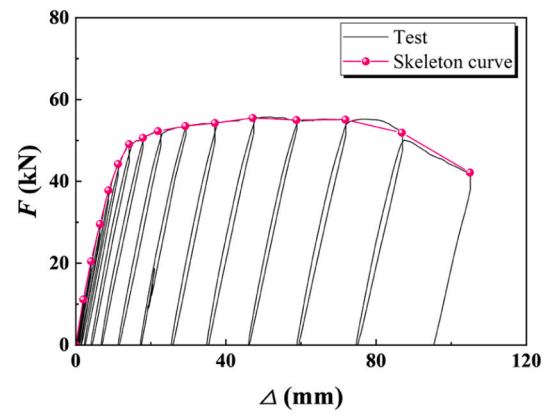


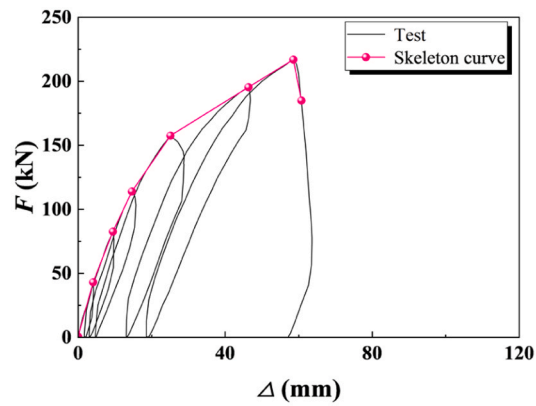
Fig. 10. Failure mode of specimen CT-3.



(a) Specimen CT-1



(b) Specimen CT-2



(c) Specimen CT-3

Fig. 11. Load-displacement curve.

3.3. Prestressing force curves

Under combined compression-bending-shear-torsion loading, the bending moment, shear force, and torque experienced by the specimen are all derived from the same external load. The global prestressing force in the specimens exhibited specific variation trends during the loading process, as shown in Fig. 12. At the initial loading stage, the overall prestress remained nearly constant in both specimens CT-1 and CT-2, primarily because the bending moment transmitted through the horizontal joint was relatively small, and no cracking had yet occurred. As the loading progressed, the overall prestress in both specimens gradually increased, this phenomenon is primarily attributed to the separation of the horizontal joint interface between the upper and lower tower sections following joint cracking. This trend is consistent with the variation in prestress observed in specimens subjected to pure compression-bending loading. Similarly, the overall prestress of specimen CT-3 remained nearly constant during the initial stage of loading, but gradually decreased in the later stages. This behaviour is primarily attributed to the absence of significant cracking at the horizontal joint, owing to the large size of the specimen. Instead, concrete crushing occurred in the compression zone of the joint, leading to a localized loss of prestress on that side, which ultimately resulted in a reduction in the overall prestress.

3.4. Strain-load curves

The variations in vertical strain at the eight measurement points located above the horizontal joint exhibited a distinct pattern, as shown in Fig. 13. In Fig. 13(a), measuring points 1 and 8 experienced sequential failure, which was consistent with the experimental observations: concrete crushing initially occurred at point 1 and subsequently propagated to point 8. As shown in Fig. 13(b), the compressive strain at measuring point 1 increased initially and then approached zero. This was attributed to local concrete spalling at this location under increasing load, after which the crushed concrete no longer participated in load transfer. In the later loading stage, measuring points 2 and 3 exhibited tensile strain, which can be explained by the crushing of concrete beneath the horizontal joint in the corresponding regions. The absence of supporting concrete led to load redistribution, resulting in tensile stresses at these points. Similarly, Fig. 13(c) shows that the damage sequence at measuring points 1 and 8 was consistent with the test observations. Measuring point 6 initially underwent compression but later exhibited unloading behaviour. This was primarily due to torsional deformation of the specimen, which induced relative displacement between the upper and lower tower segments. As a result, the edge region of the horizontal

joint beneath this point lost support, causing the measuring point to cease experiencing compressive stress.

3.5. Displacement increment curves

During the loading process, the vertical displacement responses at the four measurement points on the horizontal joint exhibited a distinct and regular pattern, as shown in Fig. 14. Because the applied load was introduced through the contraction of the actuator, the compression zone was located on the side closer to the actuator, and the displacement evolution corresponded well with this expectation. From the comparison, it can be observed that specimen CT-2 exhibits the smallest vertical displacement variation at the horizontal joint. Furthermore, during the intermediate and later loading stages, the vertical displacements at all measurement points increased slowly with the applied loading displacement. This behaviour may be attributed to the fact that, as loading progresses, once only a small area of the joint disengaged, the friction at the horizontal joint section was insufficient to resist the torsion-induced shear stress, resulting in torsion of the upper tower segment. Some measurement points of specimens CT-1 and CT-3 show abrupt changes during the final loading stage. This phenomenon can be ascribed to the local crushing of concrete in the compression zone of the horizontal joint, which occurred first as the specimens approach the ultimate state, thereby affected the readings at the corresponding measurement points.

4. FE modelling

In this chapter, numerical simulations of the concrete tower under combined compression-bending-shear-torsion loading were performed using the FE software ABAQUS [23]. The analysis accounted for both geometric and material nonlinearities, while the experimental loading system and procedures were appropriately simplified to enhance computational efficiency.

4.1. Material properties

The compressive behavior of concrete was modelled using the Mander confined concrete stress-strain relationship [24], considering only the confinement provided by outer stirrups. The tensile response was simulated with the fracture energy approach recommended in the CEB-FIP MC90 [25], and the axial tensile strength was determined following the provisions of GB 50010-2010 [26]. Reinforcing steel was represented by an ideal elastic-plastic model, whereas steel supports and bolts were modelled as linear elastic materials. prestressed tendons were simulated using a bilinear model in accordance with BS EN 1992-1-1:2014 [27], with a thermal expansion coefficient of 1.0×10^{-5} .

4.2. Elements and meshing

Concrete was modelled using eight-node reduced-integration solid elements (C3D8R). Ordinary reinforcing steel, bolts, and prestressed tendons - assumed to carry only axial forces - were represented by truss elements (T3D2), while the foundation was simulated using shell elements (S4R). To balance computational accuracy and efficiency, the transition section and ground components were assigned elastic properties with a mesh size of 120×120 mm. Concrete solids and reinforcing steel trusses were meshed at 30×30 mm, the foundation at 120×120 mm, and each prestressed tendon was modelled with a single element. Details of the meshing strategy is shown in Fig. 15.

4.3. Contact interaction and boundary conditions

The bond between ordinary reinforcing steel and concrete was simulated using the embedded region technique. Surface-to-surface

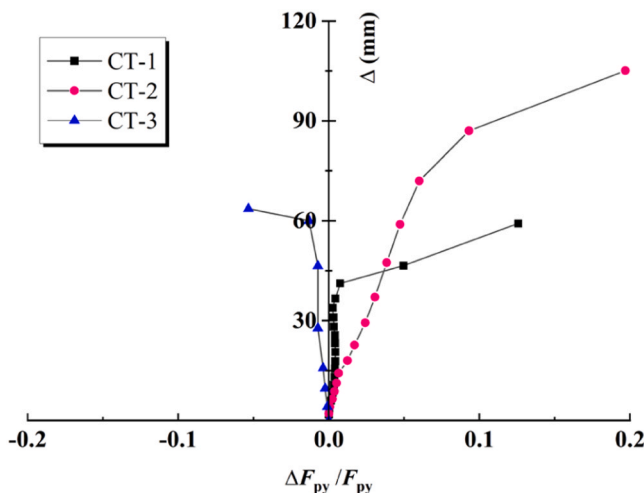


Fig. 12. Prestressing force increment-displacement curve.

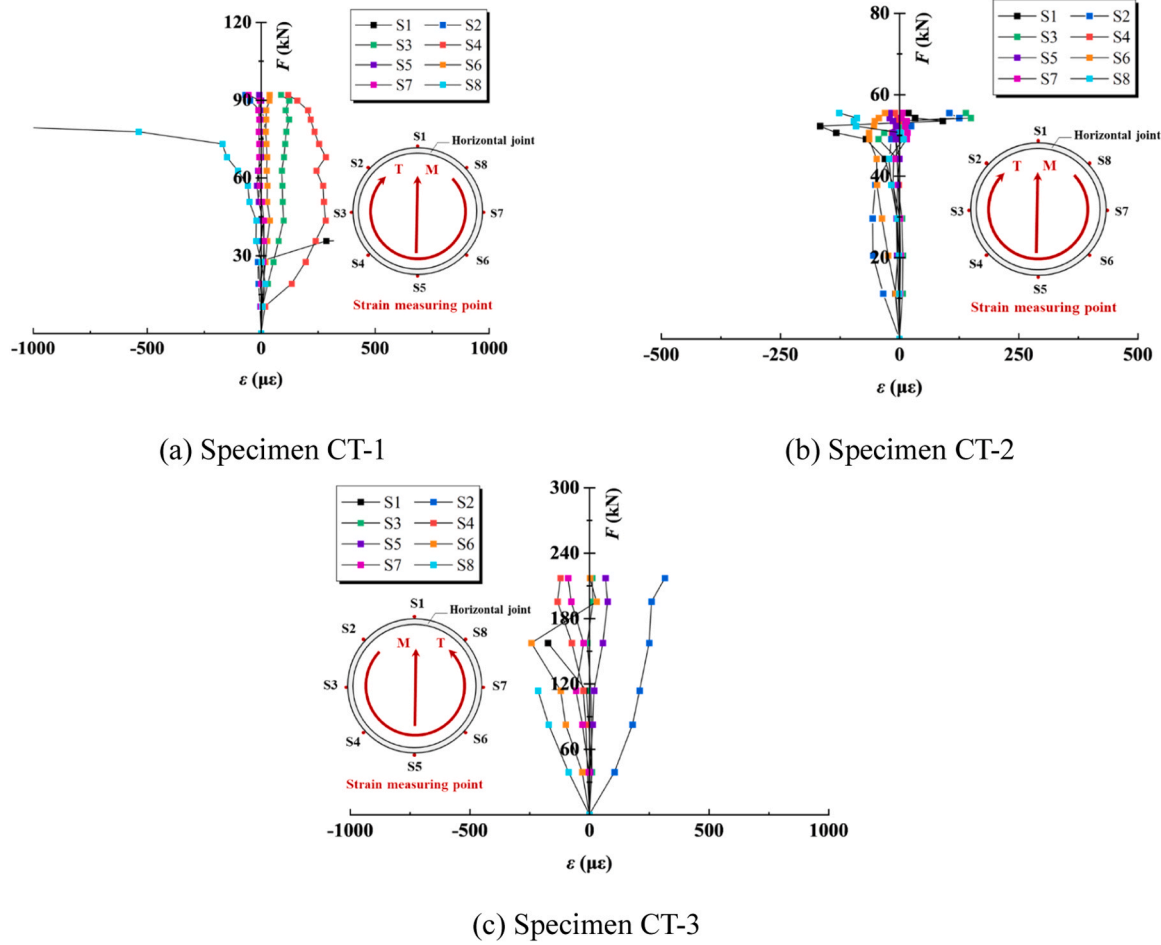


Fig. 13. Strain increment-load curves.

contact was defined for: (1) the interface between the transition tower section and the upper tower specimen, (2) the interface between the lower tower specimen and the foundation, (3) the contact surface between the foundation and the ground, and (4) the horizontal joint region. A friction coefficient of 0.5 [28] was assigned in the tangential direction for all contact pairs, while hard contact was adopted in the normal direction. A reference point RP-1 was created at the mid-height of the loading beam along the tower axis and kinematically coupled to the bolt holes at the upper end of the transition tower section. A prestress anchorage reference point RP-2 was positioned at the elevation of the upper surface of the loading beam, as shown in Fig. 15, and was kinematically coupled to both the upper ends of the prestressed tendons and the upper surface of the transition tower section. A rigid rod was installed at the model top, with a rigid body constraint applied to a reference point RP-4, as shown in Fig. 15 and coupled to the transition tower section. The lower ends of the prestressed tendons were coupled to the reserved holes in the foundation. The ground was modelled as a rigid body with fixed boundary conditions, and fixed constraints were also applied at the bolt hole locations of the foundation component. To achieve a balance between modelling accuracy and computational cost, the loading beam was represented by a rigid rod. This decision was supported by preliminary FE simulations and experimental observations, which confirmed that the beam's deformations were sufficiently small to be disregarded under the specific load cases considered.

4.4. Model validation

A comparison between the load-displacement curves obtained from the FE validation model and those measured in the experiment is

presented in Fig. 16. The numerical results exhibit close agreement with the experimental data, confirming the high accuracy of the FE model in reproducing the load-carrying capacity of the specimen.

A comparison of the peak bearing capacities obtained from each experiment and the corresponding FE simulation is presented in Table 2. The results show that the FE predictions deviate from the experimental values by less than 5%, demonstrating the capability of the FE model to accurately capture the load-carrying capacity of the specimen under combined loading conditions.

The failure modes of the specimens obtained from both the experiments and FE simulations are shown in Fig. 17. The failure locations and characteristics from the two approaches exhibit good agreement, confirming that the FE model developed in this study can reliably predict the load-carrying capacity of horizontal joints in concrete towers subjected to combined loading. Notably, issues such as deformation of the loading devices and clearances between loading devices may arise during the experiments, which cannot be avoided or simulated in the numerical model, leading to some discrepancy in stiffness between the simulation model and the test specimens.

5. Parametric study

To examine the interaction of bending, torsion, and shear for the horizontal joint in concrete towers under combined loading, a parametric analysis was carried out. The study was conducted using the refined FE model, in which material properties and reinforcement ratios were kept consistent, while the initial precompression ratio of all specimens was uniformly set to 0.1.

Different torsional effects were introduced by varying the torque arm

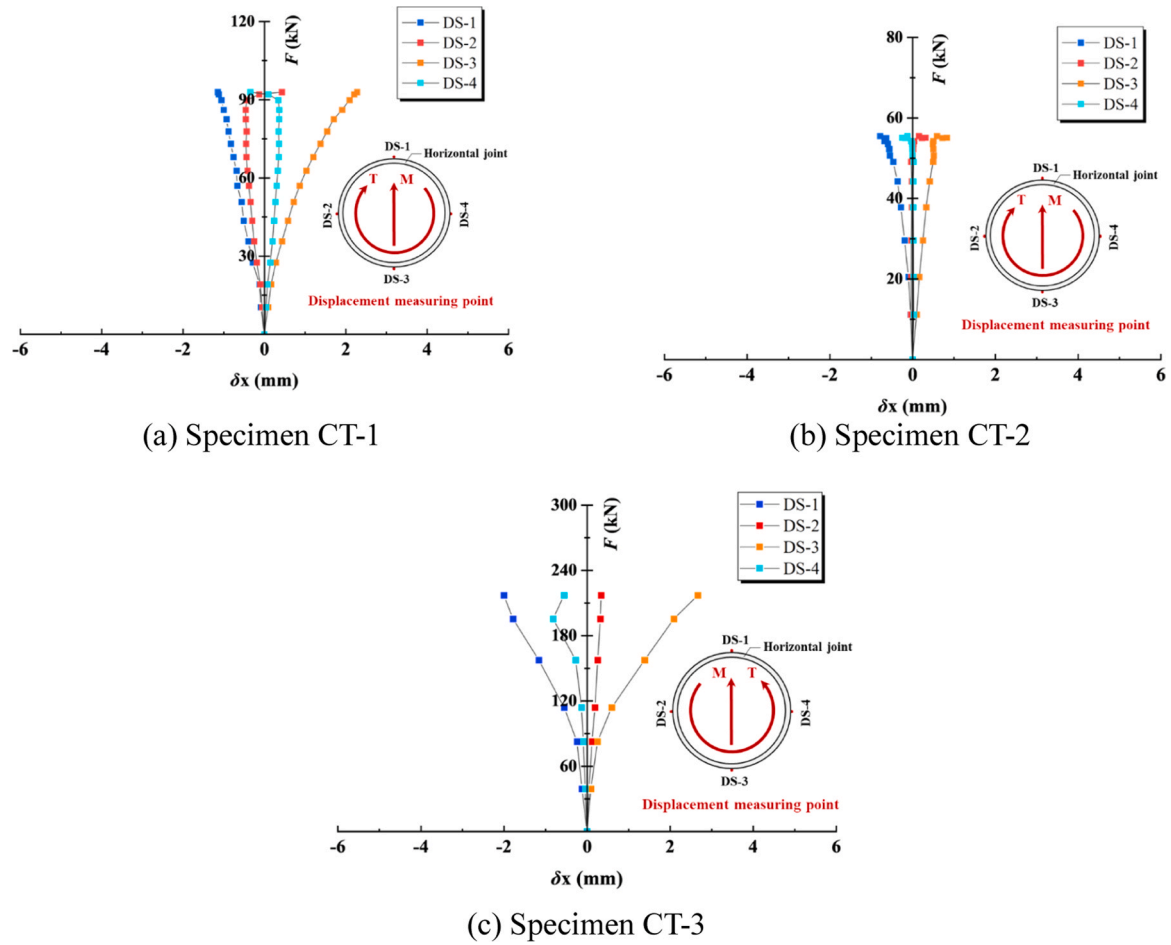
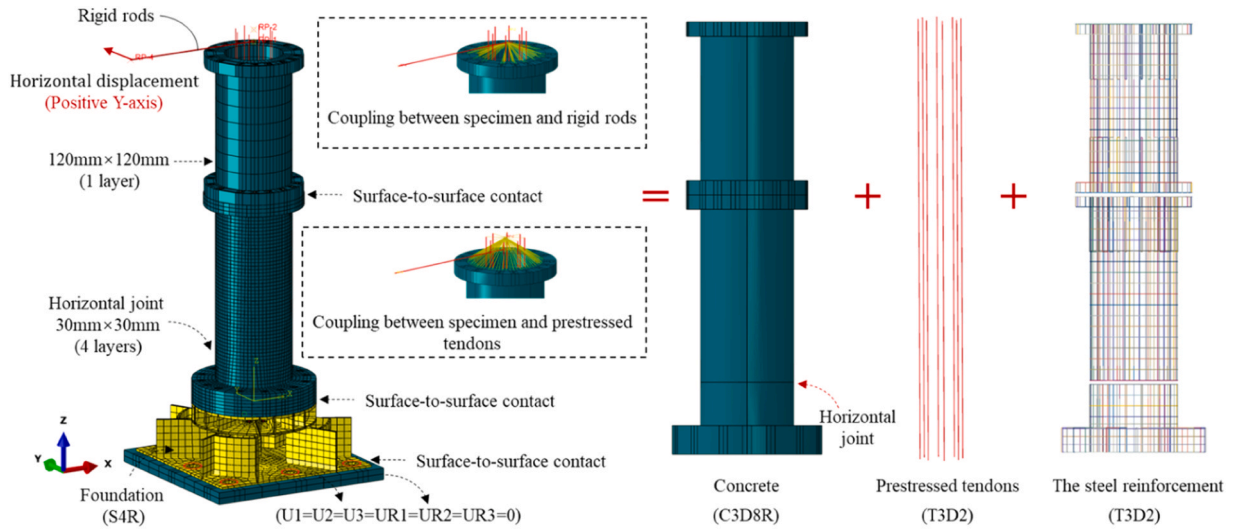


Fig. 14. Vertical displacement-loading displacement curves of horizontal joints.



length. The ratio of torque arm length to specimen diameter, referred to as the torsion-to-diameter ratio (L/D), was adjusted between 0.5 and 2.5, while all other specimen parameters were kept constant. The corresponding results are presented in Fig. 18. As the torsion-to-diameter ratio increases, the torsional effect induced by the horizontal load is amplified linearly, thereby increasing the circumferential shear stress component acting on the horizontal joints of the concrete tower.

Assuming that the total shear resistance capacity of the horizontal joints remains constant, the external shear force that the structure can withstand must be reduced linearly to accommodate this increased shear stress. Consequently, the bearing capacity of the horizontal joints shows an approximately linear decrease with the increase in the torsion-to-diameter ratio.

The theoretical torsional and flexural capacities of each model were

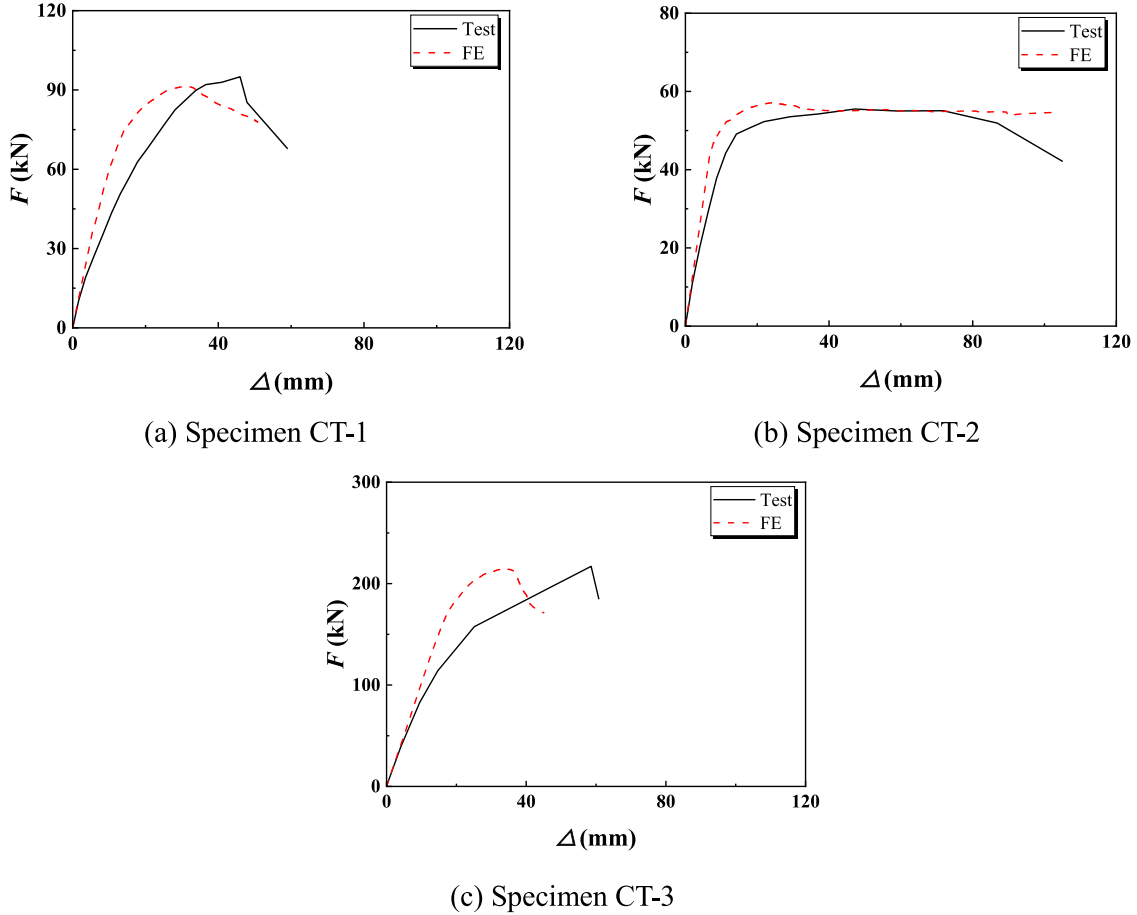


Fig. 16. Comparison of skeleton curves between FE model and test.

Table 2

Comparison of bearing capacity between experimental and FE results (kN).

Specimens	CT-1	CT-2	CT-3
Test	94	55	217
FE	91.3	57.1	214.5
Test/FE	1.03	0.96	1.01

obtained by substituting the corresponding parameters into the torsional resistance formulation for horizontal joints in concrete towers specified in *DIN EN 1992-1-1:2011-01* [17] and the flexural capacity method proposed by Tan et al. [21], respectively. The theoretical shear capacity was calculated based on Coulomb's friction law. Subsequently, the actual torque, bending moment, and shear force acting on the horizontal joint of each model were obtained by combining the geometric properties with the FE simulation results. The corresponding ratios of measured to theoretical values were then derived, defined as the moment ratio (M/M_u), torsion ratio (T/T_u), and shear ratio (V/V_u). The relationships between torsion ratio and moment ratio, shear ratio and moment ratio, and the combined torsion-shear ratio and moment ratio are presented in Figs. 19–21, respectively.

Based on the results of the parametric analysis, it can be concluded that the load-carrying capacity of the horizontal joint decreases progressively with increasing torsion-to-diameter ratio. The torsion ratio and moment ratio, as well as the combined torsion-shear ratio and moment ratio, exhibit an inverse correlation, whereas the shear ratio and moment ratio show an approximately linear relationship. These findings suggest that bending and torsion exert a mutual weakening effect on the load-carrying capacity of the horizontal joint for wind

turbine towers.

6. Evaluation method

6.1. Existing design methods

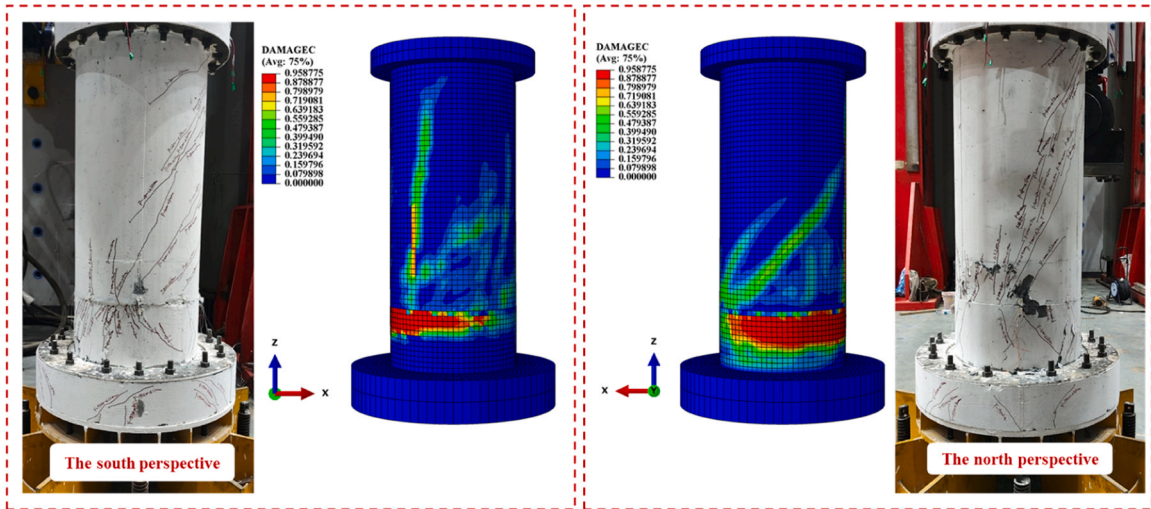
In current design practices, there's no explicit method exists for evaluating the overall load-carrying capacity of tower horizontal joints subjected to combined compression-bending-shear-torsion loading. However, numerous analytical models are available for estimating the load-carrying capacity of horizontal joints under single or simplified loading conditions. Section 6.2.3 in *T/CEC 5008* [15] provides an analytical approach for evaluating the torsional capacity of the horizontal joint in concrete towers, as given in Eq. (2). In this method, the torsional resistance is primarily attributed to the frictional force developed along the section interface, while the influence of shear-induced weakening on the overall capacity is also considered.

$$2V + \frac{T_e}{r} \leq \mu N \quad (2)$$

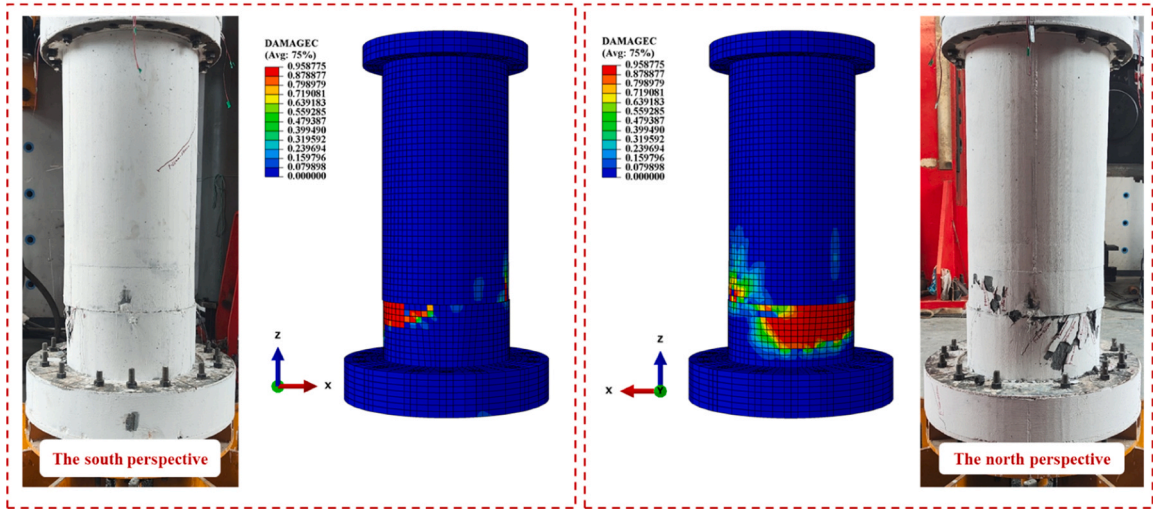
where V represents the design shear force of the horizontal section; μ indicates the interface friction coefficient of the tower horizontal joint.

DIN EN 1992-1-1 [17] specifies a calculation method for evaluating the torsional capacity of thin-walled sections, as given in Eq. (3). Although this method is applicable to the analysis of homogeneous continuous members, the prestress generates significant frictional forces at the horizontal joints. Therefore, this approach is employed for the indirect evaluation of the bearing capacity of the horizontal joints.

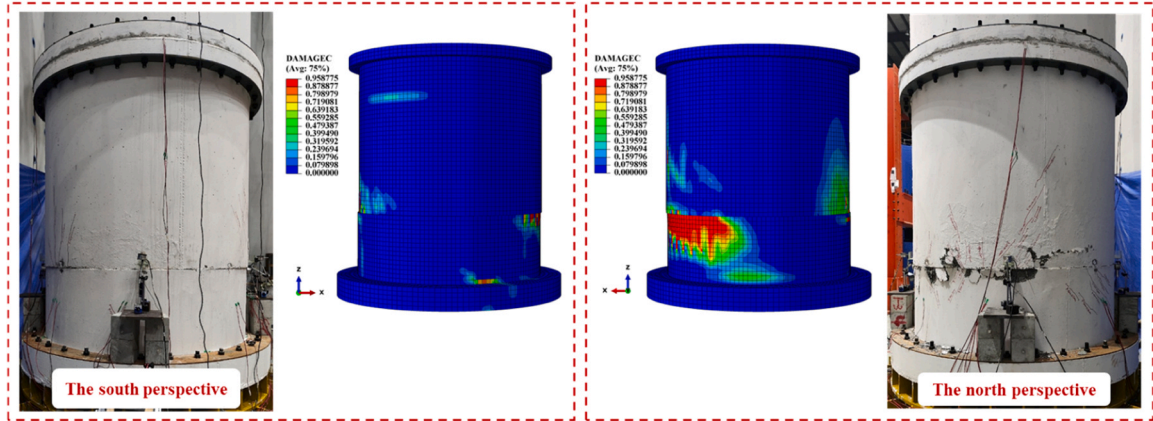
$$T_e = 2A_k t_{ef,i} \tau_{t,i} \quad (3)$$



(a) Specimen CT-1



(b) Specimen CT-2



(c) Specimen CT-3

Fig. 17. Comparison of failure modes between FE model and test.

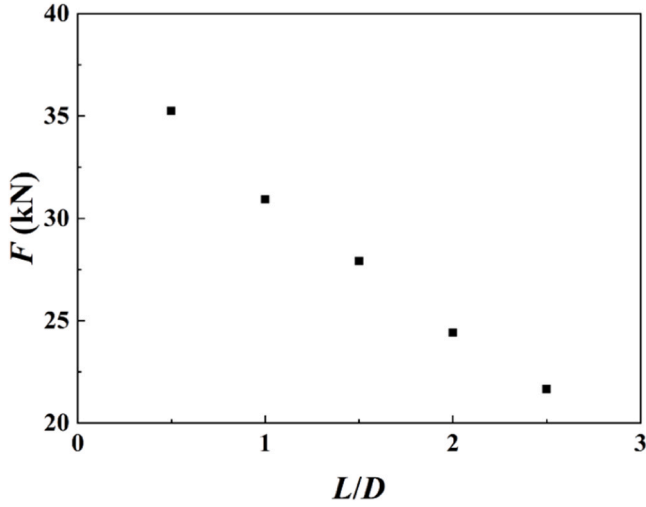
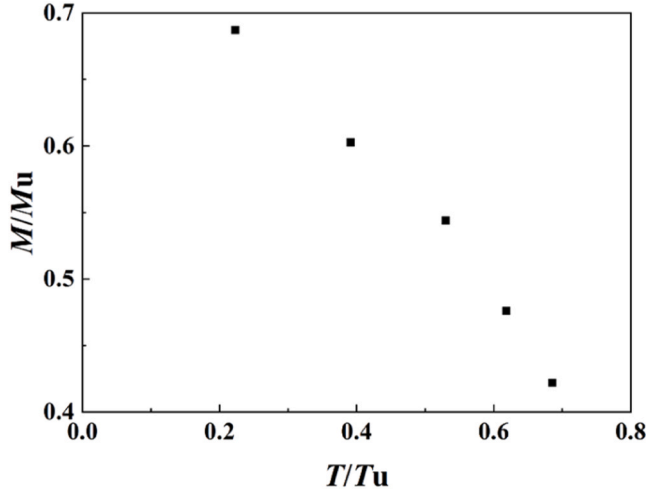
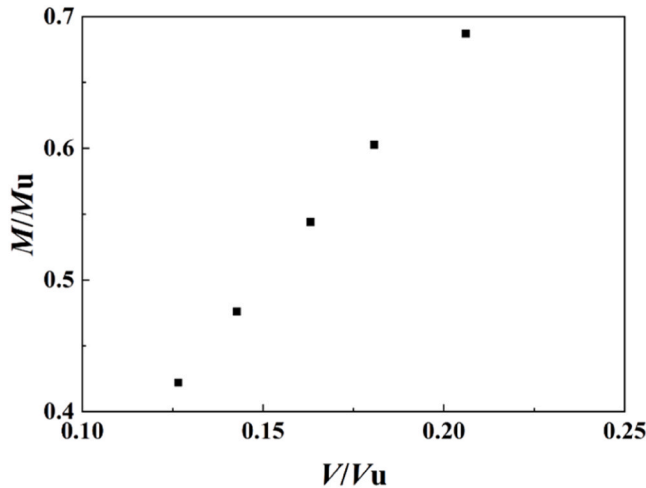
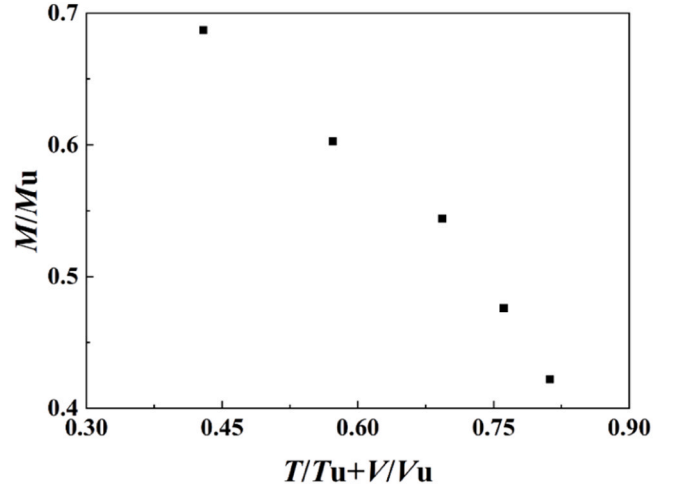


Fig. 18. Relationship between torsional-diameter ratio and load.

Fig. 19. Relationship between T/T_u and M/M_u .Fig. 20. Relationship between V/V_u and M/M_u .

where A_k represents the area enclosed by the centre line of the circular section; $t_{ef,i}$ indicates the wall thickness of the concrete tower; $\tau_{t,i}$ denotes the torsional shear stress of the section.

Fig. 21. Relationship between $T/T_u + V/V_u$ and M/M_u .

The Chinese design code GB 50010 [26] provides a calculation approach for determining the flexural capacity of circular-section members, as expressed in Eq. (4). This method is applicable to prestressed concrete members with circular cross-sections, in which the flexural resistance is governed by the combined contribution of the tensile capacity of prestressed tendons on the tension side and the compressive strength of concrete on the compression side. Although the concrete wind turbine towers are typically reinforced concrete components rather than continuous prestressed members, their reinforcement ratios are primarily governed by detailing requirements. Therefore, this calculation method remains applicable in practical engineering design.

$$M_e = \alpha_1 f_c A (r_1 + r_2) \frac{\sin \pi \alpha}{2\pi} + f_{py} A_p r_p \frac{\sin \pi \alpha}{\pi} + (f_{py} - \sigma_{p0}) A_p r_p \frac{\sin \pi \alpha_t}{\pi} \quad (4)$$

where α_1 is the ratio of the stress value of the rectangular stress block for compressed zone concrete to the design value of axial compressive strength of concrete; A is the area of circular section; r_1 and r_2 are the inner and outer radii of the section; α is the ratio of compression zone area to the total section area, calculated by Eq. (5); f_{py} and f'_{py} are the design values of the tensile and compressive strengths of the prestressed tendons; A_p is the total cross-sectional area of the prestressed tendons; r_p is the distribution radius of the prestressed tendons; σ_{p0} is the tensile stress of prestressed tendons after tensioning; α_t is the ratio of the cross-sectional area of tensile reinforcement to the total cross-sectional area of all reinforcement, calculated by Eq. (6). When α exceeds $2/3$, α_t is taken as 0.

$$N = \alpha \alpha_1 f_c A - \sigma_{p0} A_p + \alpha f'_{py} A_p - \alpha_t (f_{py} - \sigma_{p0}) A_p \quad (5)$$

where N is the design value of the axial force.

$$\alpha_t = 1 - 1.5\alpha \quad (6)$$

Based on the modelling conditions of the validated FE model, selected parameters were adjusted, and extensive numerical simulations were conducted. The adjusted parameters and their corresponding ranges were as follows: concrete compressive strength from C60 to C80; diameter-to-thickness ratios between 10 and 30; initial precompression ratios from 0.10 to 0.35; and the bending-to-torsion ratio from 1 to 8, defined as the ratio of the bending moment arm to the torque arm, was achieved by adjusting the torsion-to-diameter ratio within the range of 0.5–4.0. These configurations enabled a total of 125 FE models, forming the basis for the subsequent derivation of the evaluation equation.

The bearing capacity (F) of each model was predicted using the three aforementioned analytical methods, and the calculated results were compared with the simulated bearing capacity (F_0), as shown in Fig. 22.

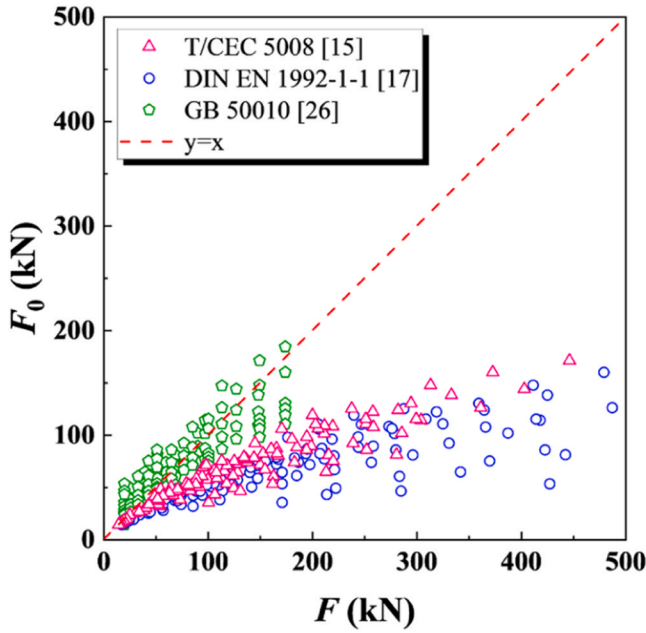


Fig. 22. Verification of existing calculation methods.

The comparison reveals that none of the three methods can accurately predict the load-carrying capacity of horizontal joints under combined loading conditions. Among them, the formulations specified in *T/CEC 5008* [15] and *DIN EN 1992-1-1* [17] significantly overestimate the bearing capacity, indicating unsafe predictions, whereas the method in *GB 50010* [26] exhibits large variability and limited reliability.

6.2. Theoretical model

For onshore wind turbine concrete towers, the primary loads include vertical loads and horizontal components induced by wind action, respectively. The vertical load comprises the prestressing force, the self-weight of the tower, and the weight of the upper nacelle and rotor. Under wind excitation, the tower is subjected to lateral forces that generate bending stresses and may also induce torsional and shear effects. Under extreme loading conditions, the horizontal joints of the tower experience a combined stress state of compression, bending,

shear, and torsion. Separation is likely to take place at the tension-side joint, which consequently leads to stress redistribution within the compression-side section, as illustrated in Fig. 23.

To ensure the safety of the horizontal joint connection, this study proposes a method for evaluating the ultimate load-carrying capacity of horizontal joints in concrete wind turbine towers under combined compression-bending-shear-torsion loading. The proposed method builds upon the fundamental form of the core interaction equation for evaluating the load-carrying capacity of members under combined loading. It integrates the results of the parametric analyses, while provides a comprehensive representation of the interaction among multiple stress components and is expressed in Eq. (7).

$$\left(\frac{M}{M_u}\right)^\gamma + \left(\frac{T}{T_u} + \frac{V}{V_u}\right)^\theta \leq 1 \quad (7)$$

where M , T , and V represent the flexural, torsional, and shear capacities of the horizontal joint in concrete towers, respectively, as obtained from FE simulation results. M_u , T_u and V_u denote the corresponding capacities predicted using existing analytical methods. Specifically, the values of T_u and M_u are calculated based on the methods proposed in Reference [17] and [21], respectively.

The torsional capacity (T_u) of horizontal joint in concrete wind turbine towers is calculated according to Eq. (8):

$$T_u = \mu F_n r \quad (8)$$

where F_n indicates the prestressing force; r denotes the average radius of the horizontal joint.

The flexural capacity (M_u) of the horizontal joint, considering the combined compression-bending effect, is calculated using Eqs. (9) - (12):

$$\varphi = \alpha_{py} \left(\frac{D}{t}\right) \left(\frac{f_{ck}}{f_{ck,60}}\right), (10 \leq \frac{D}{t} \leq 50; 39 \leq f_{ck} \leq 58; 0.1 \leq \alpha_{py} \leq 0.4) \quad (9)$$

$$\beta = \frac{2.2}{(\varphi + 1)^2} + 1, (1 \leq \varphi \leq 30) \quad (10)$$

$$\alpha = \frac{F_z + \beta \sigma_{py} A_p}{\alpha_1 f_{ck} A} \quad (11)$$

$$M_u = \alpha_1 f_{ck} A r \frac{\sin \pi \alpha}{\pi} \quad (12)$$

where φ represents the characteristic coefficient; β indicates the

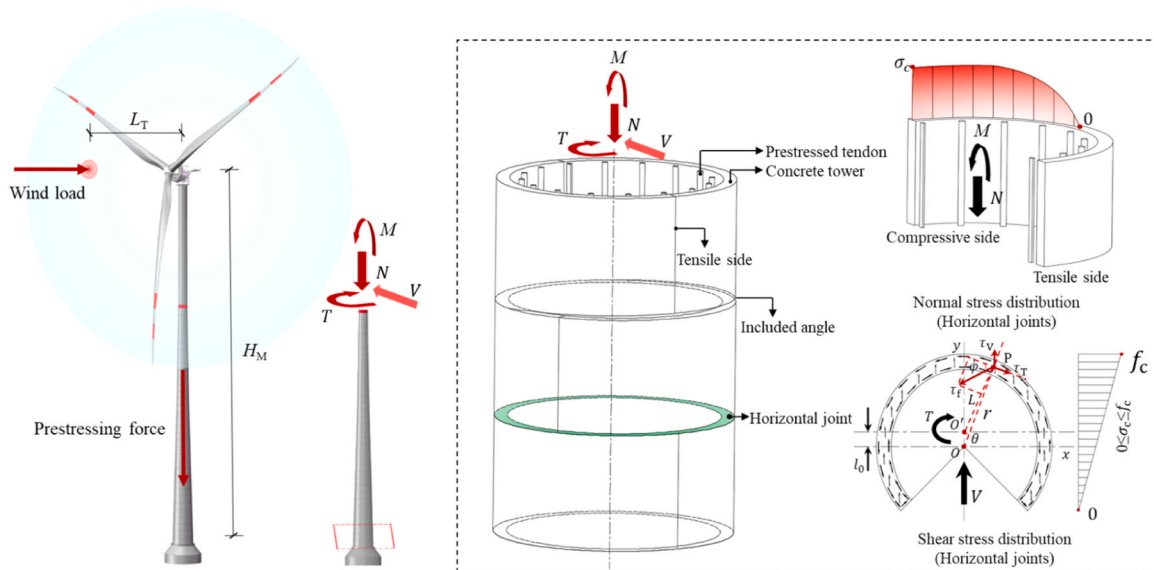


Fig. 23. The stress mode of horizontal joint under compression-bending-shear-torsion loads.

prestress increment coefficient; α denotes the half-angle coefficient of the compression zone.

The shearing capacity (V_u) of horizontal joint in concrete wind turbine towers is calculated according to Eq. (13):

$$V_u = \mu F_n \quad (13)$$

The proposed evaluation method more accurately captures the combined effects of bending moment, torsion, and shear at the horizontal joint of concrete towers under wind loads. Owing to the limited availability of experimental data, further refinement of the proposed formula and determination of its coefficients need to be carried out through extensive numerical simulations.

6.3. Calculation method

The relationship between M/M_u and $T/T_u + V/V_u$ for all FE models and test specimens is presented in Fig. 24. Here M , T , and V represent the flexural, torsional, and shear capacities of the concrete tower horizontal joint obtained from numerical simulation results, respectively, while M_u , T_u and V_u denote the corresponding capacities predicted using existing calculation methods. All results from numerical simulation and tests are presented in a single plot. Based on the previously proposed evaluation framework, Curve fitting of the complete dataset was carried out, from which the key coefficients of the proposed evaluation equation were derived. The resulting relationship yields the proposed evaluation equation for the horizontal joint load-carrying capacity of concrete towers under combined loading, as given in Eq. (14). The component parameters applicable to this formula are as follows: concrete compressive strength from C60 to C80; diameter-to-thickness ratios between 10 and 30; initial precompression ratios from 0.10 to 0.35; and the bending-to-torsion ratio from 1 to 8.

$$\left(\frac{M}{M_u}\right)^{1.14} + \left(\frac{T}{T_u} + \frac{V}{V_u}\right)^{1.09} \leq 1 \quad (14)$$

7. Conclusions

This study experimentally and numerically investigated the load-carrying capacity of horizontal joints in prefabricated concrete towers under combined compression-bending-shear-torsion loading at different prestress levels. Three scaled specimens were tested, and a validated FE model was employed to conduct extensive parametric analyses. An evaluation method for the capacity of horizontal joints under combined loading was subsequently developed. The main conclusions are as follows: (1) The torsional and shear capacities of horizontal joints in concrete towers are predominantly provided by friction at the joint interface, which is directly governed by the initial prestress level; (2) According to the small scale test results, increasing the precompression ratio from 0.15 to 0.3 leads to an 70.9% improvement in the load-carrying capacity of the specimen; (3) The prestress level influences the failure mode: bending failure occurs at lower prestress levels, whereas higher prestress levels tend to induce combined torsional-bending failure; (4) Building upon the existing calculation approaches for flexural and torsional capacities of the tower's horizontal joints, and drawing on the core interaction equation for assessing the combined load-carrying capacity of structural members, a new comprehensive evaluation model is proposed for the horizontal joints of concrete wind turbine towers, which is applicable to the following parameter ranges: concrete compressive strength from C60 to C80; diameter-to-thickness ratios between 10 and 30; initial precompression ratios from 0.10 to 0.35; and the bending-to-torsion ratio from 1 to 8. It is worth noting that although this evaluation model was established based on dry joints, epoxy in horizontal joints in actual wind turbine towers is primarily used as a levelling agent, and its filling cannot guarantee complete compaction. Overall, this evaluation model is relatively conservative and can be employed to evaluate the load-carrying capacity of horizontal joints in

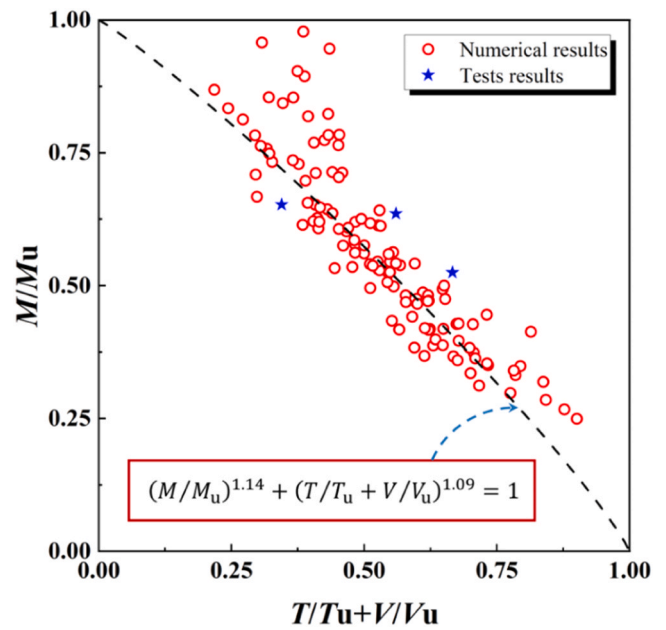


Fig. 24. The evaluation curve.

actual wind turbine towers. Furthermore, given the limited number of specimens and the exclusion of factors such as joint irregularities, further experimental and numerical investigations are recommended to extend the scope of the present study, exploring the influences of the unevenness and localized stress states of horizontal joints on their load-carrying capacities.

CRediT authorship contribution statement

Su Meini: Writing – review & editing. **Fei Deng:** Supervision, Investigation. **Qiang Li:** Supervision. **Wan-Lang Zheng:** Writing – original draft, Investigation. **Ji-Ke Tan:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Zhang Yu-Hui:** Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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